



Introduction to Node JS

1. Basics of Node JS
2. Difference from Traditional Programming
3. Scenarios for using Node JS
4. Node Setup

- Developed by Ryan Dahl in 2009
- It uses the V8 JavaScript engine, the one which is used by Google Chrome.
- Node.js applications are written in JavaScript, and can be run within the Node.js runtime
- Organizations that use Node.js
 - Paypal: for web and mobile applications
 - Netflix: for backend services
 - LinkedIn: for mobile servers
- Client side scripting languages
 - Languages like VB Script, JScript, Javascript were used in earlier days

- Features of Node JS
 - Asynchronous event-driven IO helps concurrent request handling
 - V8 JavaScript Runtime engine
 - Handling of concurrent requests
 - The Node.js library uses JavaScript
 - Cross-platform framework
 - Lightweight framework
 - Open source framework
 - Node.js is an open-source framework under MIT license. (MIT license is a free software license originating at the Massachusetts Institute of Technology (MIT).)

- **Working of traditional web servers:**
 - Each request is handled by a dedicated thread from the thread pool
 - Dedicated thread executes a particular request and returns only after completing the execution and returns a response
- **Working of NodeJS Server:**
 - Application code runs in a single thread
 - Node.js uses worker threads for performing CPU-intensive tasks.
 - Main thread need not wait for request to be completely processed and reponse to be returnde

- **Scenarios where Node JS is preferred:**
 - lightweight, high traffic, data-intensive but low processing or computation application
 - Applications that use object db.
 - Applications when the client doesn't need firm confirmation
 - Queued Input
 - Real-time web solution

- **Scenarios where Node JS should not be used**
 - A CPU-Heavy Application:
 - A Simple CRUD (or HTML) Application
 - A Relational Database-Backed Server-Side App

- Node.js development environment can be setup on any platform
- NPM (Node Package Manager) is included in Node.js installation
- Steps to follow
 - Download the Installer preferable LTS version
 - Install Node.js by simply double clicking the installer.
Features will be installed by default:
 - i. Node.js runtime
 - ii. Npm package manager
 - iii. Modification of Path
 - Verify installation by getting display version

- Executing first application
 - Step 1 : Create a JavaScript File having .js as the extension
 - Step 2 : Run node Application using node command
- REPL terminal
 - It is a built-in interactive environment.
 - It is used for testing JavaScript code, execute statements, so that result can be seen immediately.